Recitation 10

31-3-26

(Boosting)

Age	Income	w_i	credit_score	Target (loan approved)		
25	40	0.2	650	0.14	Yes → +1	0.125 +1
30	60	0.2	700	0.14	Yes → +1	0.125 +1
45	35	0.2	600	0.14	No → -1	0.125 -1
50	30	0.2	580	0.14	No → -1	0.125 -1
28	55	0.2	720	0.14	Yes → +1	0.5 +1

decision stump Many

↓

AdaBoost

1) Initialize weights: $w_i = \frac{1}{N} = \frac{1}{5} = 0.2$

2) decision stump

Age $< 40 \rightarrow$ Yes $(+1)$

$\geq 40 \rightarrow$ No (-1)

3) weighted error

$$\epsilon_t = \sum_{i=1}^{N} w_i \cdot I(y_i \neq h_t(x_i))$$

I → Indicator function

y_i ↓ Actual, $h_t(x_i)$ ↓ predic

$$\Rightarrow [0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0] = 0$$

classifier at $0.2 \times 1 = 0.2$

4) model error

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right) = \frac{1}{2} \ln\left(\frac{1-0}{0}\right) = \infty$$

$$= \frac{1}{2} \ln\left(\frac{1-0.2}{0.2}\right) =$$

$$\Rightarrow 0.693$$

5) update weighted:

$$w_i^{new} \rightarrow w_i e^{-y_i \alpha_1 h_1(x_i)}$$

$$\downarrow$$

$$e^{-\infty} = 0.$$

6) prediction

Model will stop training (x_i)

$$H(x) = h_1(x_i)$$

$$0.2 \times e^{-[1 \times 0.693 \times 1]}$$

$$0.2 \times e^{-[1 \times 0.693 \times -1]}$$

Normalized wts $= \frac{w_i}{\sum w_i} = \frac{0.1}{0.8} \Rightarrow 0.125$

$0.2 \rightarrow 0.125$

(wts decreasing) $\frac{0.4}{0.8} \Rightarrow 0.5 \rightarrow$ last one.

$0.2 \rightarrow 0.5$

(increasing)

7) Final prediction

$$H(x) = \text{sign}\left(\sum_{t=1}^{T} \alpha_t h_t(x_i)\right)$$

$$= \text{sign}\left[\alpha_1 h_1(x_1) + \alpha_2 h_2(x_2) + \cdots\right]$$